A 1.5 V, 50 mA PMOS Low-Dropout Regulator in 0.18 μm CMOS

Part 2: Hand calculation of W/L from the specification, and ngspice verification

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Abstract

This note sizes a PMOS low-dropout regulator from a written specification and checks it in ngspice. Section 2 extracts square-law parameters from the 0.18 μm BSIM3 model. Section 3 computes every W/L, bias current and resistor step by step. Section 4 verifies the design, reports three iterations, explains each gap between hand calculation and simulation, and lists the corner results. The final design meets every specification at the nominal point. **All equation, lemma, proposition and corollary numbers refer to Part 1** (`ldo_theory_proofs.pdf`), where each formula is proved; tables, figures and sections refer to this note.

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1 Specification and architecture

Table 1: Specification. Typical model, $T = 27\,^{\circ}\mathrm{C}$ unless stated. The last column is the simulated value of the final design (v3, Section 4).

Quantity	Condition	Spec	Simulated (v3)
Input voltage V_{IN}		1.7–1.9 V (nom. 1.8 V)	–
Output voltage V_{OUT}		1.50 V	1.4954–1.5053 V
Load current I_L		0.1–50 mA	–
Dropout voltage	$I_L = 50$ mA, V_{OUT} within 1 %	≤ 200 mV	141 mV
Output capacitor	ceramic	$C_L = 2.2\,\mu$F, $R_{ESR} = 0$–$50\,\Omega$	–
Phase margin	all loads, all R_{ESR} in range	$\geq 45^{\circ}$	60.0° (worst, incl. corners)
PSRR	1 kHz, $I_L = 50$ mA, $V_{IN} = 1.8$ V	≥ 30 dB	31.0 dB
Load regulation	I_L: 0.1→50 mA	$\lvert\Delta V_{OUT}\rvert \leq 15$ mV	9.9 mV
Line regulation	V_{IN}: 1.7→1.9 V	≤ 35 mV/V	29.9 mV/V (worst)
Quiescent current	excl. I_L, incl. divider	$\leq 100\,\mu$A	84.7 μA
Load transient	0.1↔50 mA, 1 μs edges	$\lvert\Delta V_{OUT}\rvert \leq 50$ mV	9.9 mV

Process and tools. The transistor model is the 0.18 μm BSIM3v3.1 file `cmos180.lib` (1.8 V devices, $t_{ox} = 4.1$ nm) from the analog-IC coursework. Simulations use ngspice-46 in batch mode, driven by Python scripts; every number in this note is read from the script outputs. The reference voltage $V_{REF} = 0.6$ V and the bias current $I_{REF} = 5\,\mu$A are ideal sources (a bandgap is outside the scope of this note).

Architecture (Fig. 1).

- *Pass device* M_P: a PMOS. A PMOS can work with $V_{IN} - V_{OUT}$ down to its saturation voltage, which is what makes a low dropout possible.
- *Feedback divider* R_1, R_2: it feeds back the fraction $\beta = R_2/(R_1 + R_2)$ of V_{OUT}.
- *Error amplifier* (EA) M_1–M_5: a five-transistor OTA with a PMOS input pair, which works at $V_{REF} = 0.6$ V. An NMOS pair would need its input common mode above $V_{GS} + V_{DS,sat} \approx 0.75$ V.
- *Buffer* M_B: a PMOS source follower with its bulk tied to its source, biased by M_{BS}. It drives the large gate capacitance of M_P from a low impedance (Part 1, Section 4.2).
- *Load*: the off-chip capacitor C_L with its series resistance R_{ESR}. The load current is an ideal current sink.

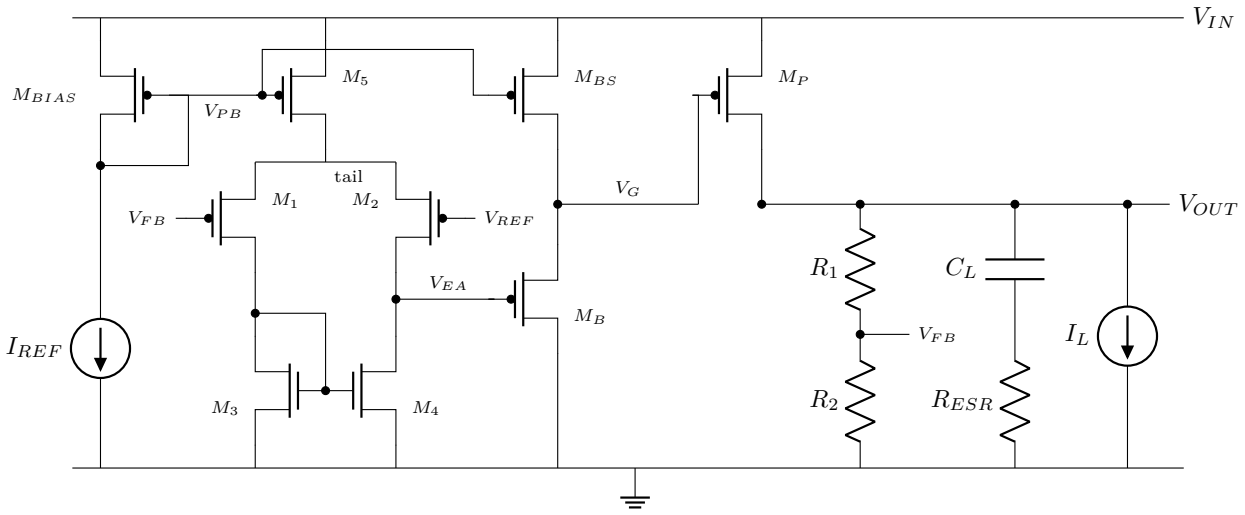


Figure 1: LDO schematic. M_1 (gate V_{FB}) sits on the diode side of the mirror, M_2 (gate V_{REF}) on the output side; Lemma 4 shows that this gives negative feedback. Bulk of every PMOS is V_{IN} except M_1, M_2 (tail node) and M_B (its source).

2 Device parameters

Extraction. For each device type and length, ngspice sweeps I_D–V_{GS} at $|V_{DS}| = 0.9$ V ($W = 10\,\mu$m). By (1), $\sqrt{I_D}$ is a straight line in V_{GS}. Its slope gives K' and its intercept gives V_{TH}; the fit uses $V_{OV} = 0.1$–0.3 V. Next, λ is the slope of I_D–V_{DS} at $V_{OV} = 0.2$ V over $|V_{DS}| = 0.6$–1.2 V, divided by I_D. Table 2 lists the results, and Fig. 2 shows the simulated g_m/I_D.

Table 2: Square-law parameters extracted from `cmos180.lib` (script `char_devices.py`).

	$L = 0.18\,\mu$m	0.5	1	2	3
NMOS V_{TH} (V)	0.479	0.459	0.428	0.410	0.404
NMOS K' (μA/V^2)	219	218	217	217	216
NMOS λ (V^{-1})	0.292	0.101	0.080	0.059	0.045
PMOS $\lvert V_{TH}\rvert$ (V)	0.465	0.431	0.412	0.399	0.394
PMOS K' (μA/V^2)	71.4	53.9	49.9	48.0	47.4
PMOS λ (V^{-1})	0.342	0.118	0.077	0.046	0.032

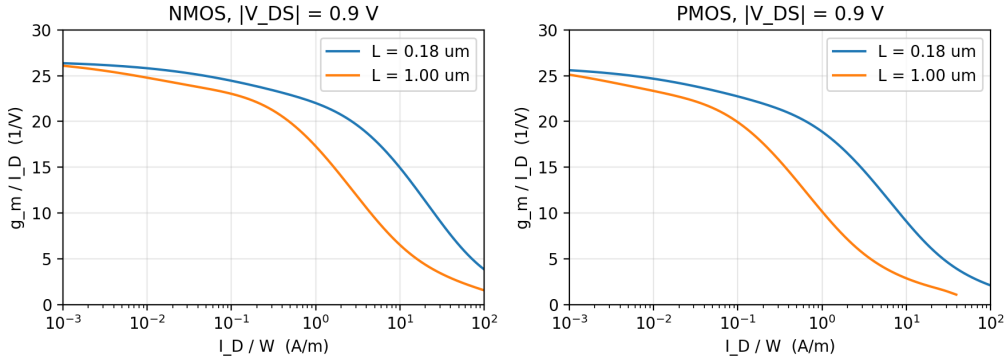


Figure 2: Simulated g_m/I_D versus current density. The square law predicts $g_m/I_D = 2/V_{OV}$, which is $20\,\mathrm{V}^{-1}$ at $V_{OV} = 0.1\,\mathrm{V}$. The real device gives about $17\,\mathrm{V}^{-1}$ there (moderate inversion) and levels off near $1/(nV_T) \approx 26\,\mathrm{V}^{-1}$ in weak inversion; this is one source of the hand-versus-simulation gap in Section 4.3.

Note on the model file. ngspice reports that the model parameters `XL` and `XW` are not recognised and ignores them. In this file $XL = 0$ and $XW = -10\,\mathrm{nm}$, so the effect is a 10 nm width change and is negligible for the widths used here.

3 Hand design from the specification

The script `design.py` carries out the steps below; each number is quoted from its output. This section describes the final design (v3). Section 4.1 explains how v1 and v2 led to it.

Step 1 – Divider. $\beta = V_{REF}/V_{OUT} = 0.40$. A divider current $I_{div} = 10\,\mu\mathrm{A}$ gives $R_2 = V_{REF}/I_{div} = 60\,\mathrm{k\Omega}$ and $R_1 = (V_{OUT} - V_{REF})/I_{div} = 90\,\mathrm{k\Omega}$.

Step 2 – Pass device. Eq. (6) with $K' = 71.4\,\mu\mathrm{A/V}^2$ (PMOS, $L = 0.18\,\mu\mathrm{m}$) and $V_{IN,nom} - V_{OUT} = 0.3\,\mathrm{V}$ (10 mV margin) gives $W/L \geq 16644$. Rounding up to 300 fingers of $10\,\mu\mathrm{m}$ gives $W = 3.0\,\mathrm{mm}$ and $W/L = 16667$. Then:

- Full load: $V_{OV} = 290\,\mathrm{mV}$, $V_{SG,P} = 0.755\,\mathrm{V}$, $g_{m,P} = 2I/V_{OV} = 0.345\,\mathrm{S}$.
- Dropout check (7) at $V_{DO} = 0.2\,\mathrm{V}$: $V_{OV} = 310\,\mathrm{mV}$, i.e. $V_{SG,P}^{DO} = 0.775\,\mathrm{V}$, which the drive chain can reach (Step 6).
- Light load (0.1 mA, weak inversion): the square law does not apply, so $V_{SG,P} = 0.366\,\mathrm{V}$ and $g_{m,P} = 2.38\,\mathrm{mS}$ are read from the simulated I_D–V_{SG} curve of one finger. Hence $\Delta V_{SG,P} = 389\,\mathrm{mV}$.
- Gate capacitance: $C_G = \frac{2}{3}WLC_{ox} + 2WC_{ov} = 6.96\,\mathrm{pF}$.

Step 3 – Required loop gain $G = \beta A_{EA} A_B$. Each specification gives a lower bound on G:

- PSRR (Proposition 19): $G \geq 10^{30/20} = 31.6$.
- Load regulation (Corollary 6b): $G \geq \Delta V_{SG,P}/15\,\mathrm{mV} = 25.9$.
- Line regulation (Corollary 6c): $G \geq 1/(35\,\mathrm{mV/V}) = 28.6$.

The binding one is the PSRR: $G \geq 31.6$, i.e. $A_{EA} A_B \geq 79$.

Step 4 – Error amplifier gain. $A_{EA} = (2/V_{OV1})/(\lambda_p + \lambda_n)$ with $V_{OV1} = 0.10\,\mathrm{V}$, $L_{12} = 1\,\mu\mathrm{m}$ and $L_{34} = 3\,\mu\mathrm{m}$: $\lambda_p + \lambda_n = 0.122\,\mathrm{V}^{-1}$, so $A_{EA} = 164$ (44.3 dB). Buffer at $I_B = 30\,\mu\mathrm{A}$: $A_B = 0.979$. Hence $G = 64.1$ (36.1 dB), which is above the requirement.

Step 5 – Unity-gain frequency and poles.

- Eq. (12) at full load: $f_u = 1.60\,\mathrm{MHz}$.
- Buffer pole: $g_{mB} = 2I_B/V_{OV,B} = 0.27\,\mathrm{mS}$, so $p_B = 6.2\,\mathrm{MHz}$.
- EA output capacitance: $C_{EA} = 192\,\mathrm{fF}$, of which M_2 contributes 139 fF. With $R_{o,EA} = 409\,\mathrm{k\Omega}$ this gives $p_{EA} = 2.02\,\mathrm{MHz}$.

- ESR zero at $R_{ESR} = 10\,\mathrm{m\Omega}$: $z_{ESR} = 7.2\,\mathrm{MHz}$.

Evaluating the full transfer function (9) numerically gives PM = 55° at full load (45° with $R_{ESR} = 0$) and 90° at light load. The hand PM with $R_{ESR} = 0$ sits right at the 45° limit, and the ESR bound (15), 45 mΩ, is slightly below the 50 mΩ end of the range. Both are pessimistic: the bound is only sufficient, and the hand G is too high (Section 4.3). The simulation (Fig. 4, right) decides.

Step 6 – EA current and buffer V_{SG}. By Proposition 16, the EA current does not move p_{EA}, so it is set by the power budget: $I_{D1} = 20\,\mu\mathrm{A}$ per side. The window (8) gives $0.522\,\mathrm{V} \le V_{SG,B} \le 0.775\,\mathrm{V}$. Its centre, 0.649 V, sets $V_{OV,B} = 0.22\,\mathrm{V}$.

Step 7 – Predictions and current budget.

- Load regulation $\Delta V_{SG,P}/G = 6.1\,\mathrm{mV}$.
- Line regulation $1/G = 15.6\,\mathrm{mV/V}$.
- $\mathrm{PSRR}_{DC} = 36.1\,\mathrm{dB}$.
- $I_Q = I_{REF} + 2I_{D1} + I_B + I_{div} = 85\,\mu\mathrm{A}$.

Table 3: Final device sizes (v3). M_{BIAS}, M_5 and M_{BS} form a mirror; the widths set the current ratios.

Device	W (μm)	L (μm)	Role / current
M_{BIAS}	5	1	diode, $I_{REF} = 5\,\mu\mathrm{A}$
M_5	40	1	tail, $40\,\mu\mathrm{A}$
M_1, M_2	81	1	input pair, $V_{OV} \approx 0.1\,\mathrm{V}$
M_3, M_4	25	3	mirror load, $V_{OV} \approx 0.15\,\mathrm{V}$
M_{BS}	30	1	buffer bias, $30\,\mu\mathrm{A}$
M_B	12	0.5	source follower, bulk = source
M_P	300×10	0.18	pass device
R_1 / R_2	90 kΩ / 60 kΩ		divider

4 Simulation

Method. The loop gain uses the break of Proposition 9. A 10^9 H inductor from the divider tap to the gate of M_1 closes the loop at DC, and a 10^3 F capacitor injects the AC test voltage at the gate. Because the gate draws no current, this measures (9) exactly for $f \gg 1/(2\pi\sqrt{LC})$. The PSRR test puts an AC source on V_{IN} with the loop closed. Line, load and dropout are DC sweeps, and the load step is a transient run with 1 μs edges.

Static output equation. Insert the simulated operating-point voltages and the simulated $A_{EA} = g_{m1}/(g_{ds2} + g_{ds4})$ into (5):

- Light load: $\varepsilon = 214\,\mathrm{mV}$, so the formula gives $V_{OUT} = 1.5055\,\mathrm{V}$. Simulation: 1.5053 V.
- Heavy load: $\varepsilon =$-183 mV, so the formula gives 1.4953 V. Simulation: 1.4954 V.

The equation therefore explains both the ±5 mV output error and its change of sign with load.

4.1 Design iterations

Table 4: Three iterations (nominal: $V_{IN} = 1.8\,\mathrm{V}$, 27 °C, $R_{ESR} = 10\,\mathrm{m\Omega}$; corner columns: $V_{IN} \in \{1.8, 1.9\}\,\mathrm{V}$, $T \in \{0, 27, 85\}\,^\circ\mathrm{C}$, both load ends). Values that miss the specification are marked with ×.

	v1	v2	v3 (final)
L_{12} / L_{34} (μm), $V_{OV,B}$ (V)	1 / 1, 0.15	2 / 2, 0.15	1 / 3, 0.22
Simulated A_{EA} at full load	76	119	99
PSRR at 1 kHz, 50 mA (dB)	28.8 ×	32.7	31.0
Line regulation, worst (mV/V)	39.1 ×	28.5	29.9
PM at 50 mA (deg)	81.4	55.8	71.1
PM, $R_{ESR} = 0$, worst corner (deg)	–	44.4 ×	60.0
PSRR at 1 kHz, worst corner (dB)	–	22.2 ×	29.8

- **v1** ($L = 1\,\mu$m everywhere) is stable, with a large margin, but its EA gain is only 76 against 127 by hand. As a result, PSRR and line regulation miss.
- **v2** doubles L for M_1–M_4. The gain rises, but by Proposition 16 p_{EA} falls (it scales as λ/L), and the phase margin with $R_{ESR} = 0$ drops to 44.4° at $V_{IN} = 1.9\,$V, 85 °C. At light load and $V_{IN} = 1.9\,$V, V_{EA} also rises to within 0.07 V of the bound $V_{REF} + |V_{TH,p}|$ of Proposition 8. M_2 then nears triode and the PSRR falls to 22.2 dB.
- **v3** applies both results.
  - The simulated operating point shows that g_{ds4}, not g_{ds2}, dominates the EA output conductance. v3 therefore lengthens only the mirror ($L_{34} = 3\,\mu$m) and keeps the wide input pair short ($L_{12} = 1\,\mu$m), so C_{EA} stays small.
  - $V_{OV,B}$ is moved to the centre of the headroom window (8).

4.2 Results of the final design

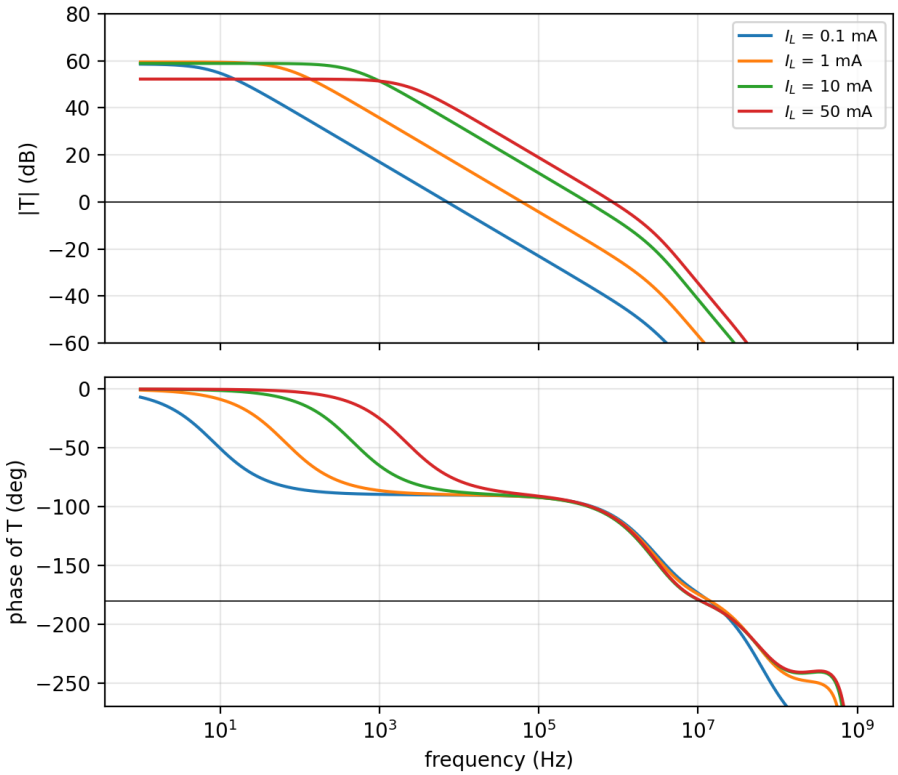


Figure 3: Loop gain T at four loads. The DC gain is about 52–60 dB, f_u moves from 7.1 kHz to 847 kHz with load (Proposition 14), and the PM is 71.1° at the worst (full) load.

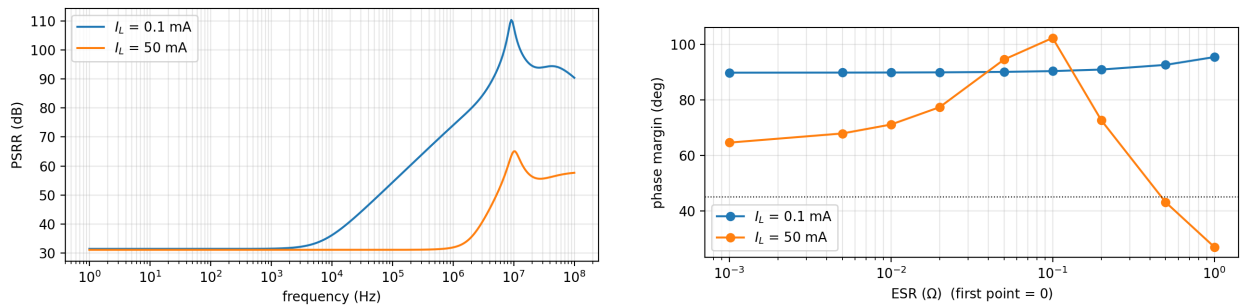


Figure 4: Left: PSRR. It is flat at about G up to the loop bandwidth, as predicted by (17); above f_u the capacitor C_L takes over. Right: phase margin versus R_{ESR}. The margin first rises (zero lead), peaks near 100 mΩ, then collapses as the plateau of Proposition 17 pushes f_u toward p_B, p_{EA}. It stays above 45° up to 200 mΩ and is 43° at 500 mΩ.

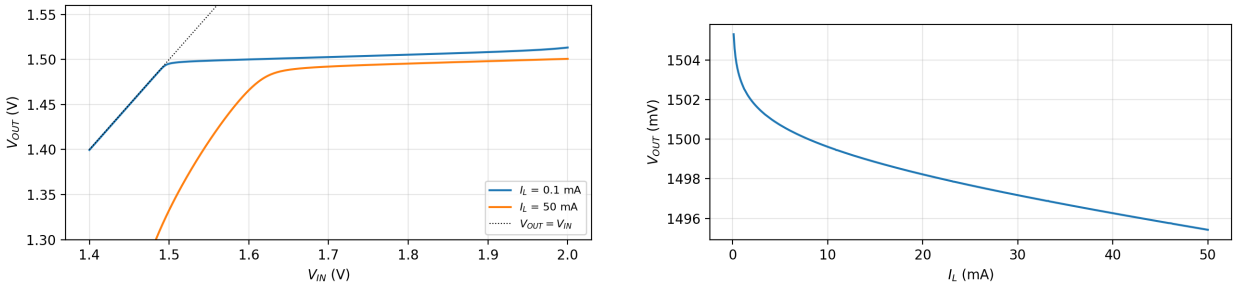


Figure 5: Left: V_{OUT} versus V_{IN}. The dropout (1 % criterion) at 50 mA is 141 mV. Right: load regulation, $\Delta V_{OUT} = 9.9\,\text{mV}$ over the full range.

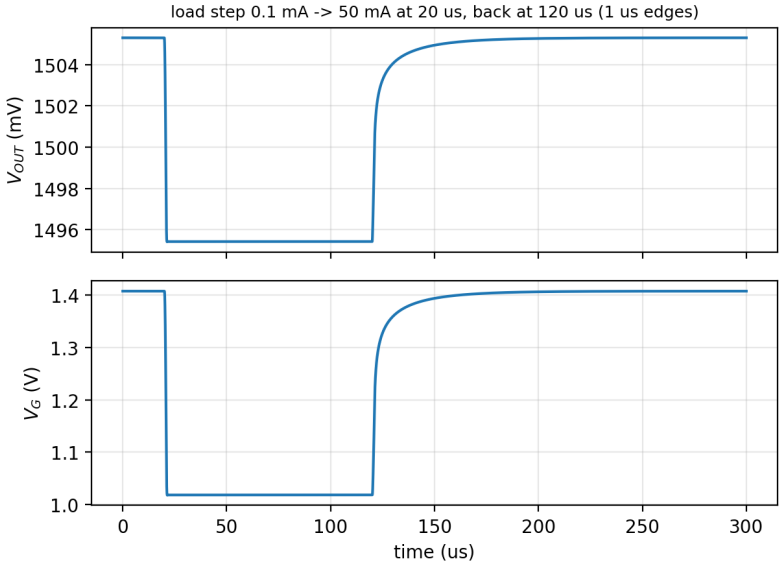


Figure 6: Load step. The step-up settles within 1 mV immediately (fast loop at full load). The release settles exponentially with a fitted $\tau = 20.9\,\mu$s, against $1/(2\pi f_{u,light}) = 22.4\,\mu$s from Proposition 21. There is no overshoot, as the single-pole model predicts. The total excursion, 9.9 mV, is almost all the static load regulation (9.9 mV).

4.3 Hand calculation versus simulation

Table 5: Final design: hand values (Section 3) and simulation. The "re-computed" column puts the simulated g_m and g_{ds} of the operating point into the same formulas.

Quantity (full load)	Hand	Re-computed	Simulated
g_{m1}/I_{D1} (V^{-1})	20	–	16.6
λ_{eff} of M_2 / M_4 (V^{-1})	0.077 / 0.045	–	0.102 / 0.069
A_{EA}	164	99	–
A_B	0.979	0.981	–
$g_{m,P}$ (S)	0.345	–	0.323
$G = \text{PSRR}_{DC}$ (dB)	36.1	31.7	31.0
f_u (kHz)	1599	903	847
PM (deg)	55	–	71
Load regulation (mV)	6.1	10.1	9.9
I_Q (μA)	85	–	84.7

The formulas hold; the gap lies in two square-law inputs.

(i) g_m/I_D. The input pair works at $V_{OV} \approx 0.08\,\text{V}$, in moderate inversion, where g_m/I_D saturates toward $1/(nV_T)$ (Fig. 2) instead of following $2/V_{OV}$. The simulated value is 16.6 against 20, about 1.6 dB of gain.

(ii) λ. The extraction was done at $V_{OV} = 0.2\,\mathrm{V}$ and $|V_{DS}| = 0.6$–$1.2\,\mathrm{V}$. In the circuit, M_2 has a smaller V_{OV} and M_4 sits at $V_{DS} = 0.37\,\mathrm{V}$, close to saturation, where g_{ds} is larger. The effective λ values are higher than the table values.

When the simulated g_m and g_{ds} are put back into the formulas, G, f_u and the load regulation match the simulation to within about 1 dB, 7 % and 0.5 mV. Because the hand G was too high, the hand f_u was too high too, and the hand PM was pessimistic.

The same mechanism affects the pass device. At full load $V_{SD} = 0.30\,\mathrm{V}$ is close to $V_{DS,sat} = 0.22\,\mathrm{V}$, so $g_{ds,P} = 30\,\mathrm{mS}$ against $\lambda I = 17\,\mathrm{mS}$. This lowers R_o and the DC loop gain (52.3 dB simulated against 62.2 dB by hand), but it does not move f_u, which is independent of R_o (Proposition 14).

4.4 Corners

The model file has no process corners, so the checks cover V_{IN} and temperature only (script `sim_extra.py`).

- *Stability.* For $V_{IN} \in \{1.7, 1.8, 1.9\}\,\mathrm{V}$, $T \in \{0, 27, 85\}\,^\circ\mathrm{C}$ and both load ends, the minimum PM is 66.3° at $R_{ESR} = 10\,\mathrm{m\Omega}$, and 60.0° with $R_{ESR} = 0$ (at 1.9 V, 85 °C).
- *Output voltage.* V_{OUT} stays within 1.4876–1.5088 V over all corners.
- *PSRR, $V_{IN} \geq 1.8$ V.* The worst value is 29.8 dB at 1.9 V, 0 °C, 0.1 mA. This is just below the 30 dB specification, which is stated at the nominal point; the cause is the remaining headroom limit of Proposition 8 at high V_{IN}.
- *PSRR, $V_{IN} = 1.7$ V.* At full load this is the dropout region, and the PSRR falls to 15.6 dB at 85 °C. Corollary 20 explains it: at the 27 °C dropout point, the simulated ratio $g_{ds,P}/g_{m,P}$ is already 0.33 (against 0.09 at 1.8 V), and it grows further when hot. Datasheets specify PSRR with V_{IN} well above $V_{OUT} + V_{DO}$ for this reason.

5 Limitations and next steps

- V_{REF} and I_{REF} are ideal. A bandgap reference and its PSRR would add to the supply path.
- Only the typical model is available: no process corners and no mismatch (Monte Carlo) analysis. Random offset of the input pair adds to ε in (5).
- The design is schematic-level only. Layout parasitics, bond-wire inductance in series with C_L, and the pass-device metal resistance are not included.
- Not covered: start-up, current limit and short-circuit protection, noise, and the soft-start of V_{REF}.
- The PSRR is limited to about G by the ground-referenced gate (Corollary 18). Possible improvements are a second EA gain stage with its own compensation, or a gate drive that tracks V_{IN}.

A Reproducing the results

The scripts, results and document sources are in the GitHub repository `https://github.com/NguyenVuQuocMinh/cmos-ldo-regulator-180nm`. The 0.18 μm model file is not redistributed there; place a BSIM3 model with `.model NMOS` and `.model PMOS` as `sim/cmos180.lib`, and set the environment variable `NGSPICE` to the ngspice console executable if it is not at the default path. Run from `scripts/`:

1. `python char_devices.py`: device parameters (Table 2, Fig. 2).
2. `python design.py`: hand design, writes `sizes_v1/v2/v3.json` and `hand.json`.
3. For each version: set `LDO_SIZES=sizes_vN.json` and `LDO_VER=vN`, then run `sim_op.py`, `sim_all.py` and `sim_extra.py`.
4. `python render.py <path-to-tectonic.exe>`: fills this document with the numbers and builds the PDF.

The netlist is generated by `ldo_netlist.py` from the sizes file. The repository folder `netlists/` holds the final circuit (v3) with its operating-point, loop-gain, PSRR and load-step test benches; each runs directly with `ngspice_con -b` once the model file is in place.